Audiophile – System Product Backlog

1. Audiophile User Stories

1.1. Select Instrument (Low Priority) | 1-point story

As a musician, I want to select the instrument I am practicing so that the system can analyze my performance based on the correct instrument settings.

1.1.1. <u>Pre-Conditions</u>

The user has accessed the web application and can view the instrument selection display. The list of supported instruments is loaded, and the system is ready to accept the user's input.

1.1.2. <u>Post-Conditions</u>

The selected instrument is stored in the session, and the system becomes configured with that instrument in mind.

1.2. ***Start Recording Audio (High Priority)*** | 2-point story

From Record Audio Epic. As a musician, I want to record or upload my audio performance so that I can receive feedback without needing a teacher present.

1.2.1. <u>Pre-Conditions</u>

An instrument has been selected. Microphone permissions are granted. The recording interface is active.

1.2.2. <u>Post-Conditions</u>

Audio recording begins successfully. The system starts capturing sound input from the microphone. A recording session is created and marked as active.

1.3. ***Stop Recording Audio (High Priority)*** | 2-point story

From Record Audio Epic. As a musician, I want to stop recording when I finish playing so that my audio file can be saved and processed.

1.3.1. <u>Pre-Conditions</u>

Recording has already started. Audio is currently being captured.

1.3.2. <u>Post-Conditions</u>

Audio recording stops immediately. The captured audio is finalized into a temporary file. The recording session is marked as complete, and the file becomes available for saving and processing.

1.4. ***Save Recorded Audio (Medium Priority)*** | 2-point story

From Record Audio Epic. As a musician, I want my recorded audio to be saved temporarily so that it can be processed for pitch detection and feedback.

1.4.1. <u>Pre-Conditions</u>

Recording has been stopped successfully. A valid temporary audio file exists.

1.4.2. <u>Post-Conditions</u>

The audio file is stored temporarily in the system session or memory. The file is ready for the audio processing stage. The user is notified that the recording has been saved successfully.

1.5. ***Upload Audio File (Medium Priority)*** | 1-point story

From Record Audio Epic. As a musician, I want to upload an audio file of my performance so I can receive feedback from a non-live recording.

1.5.1. <u>Pre-Conditions</u>

An audio file upload button is available to click and the musician has an audio file ready to be uploaded.

1.5.2. <u>Post-Conditions</u>

The audio file is saved in temporary storage and put through the processing stage.

1.6. ***Reduce Background Noise (High Priority)*** | 5-point story

From Process Audio Epic. As a musician, I want background noise in my recording to be reduced so that pitch detection is accurate and not affected by external sounds.

1.6.1. <u>Pre-Conditions</u>

A valid recorded or uploaded audio file exists. The recording session has been saved. The audio processing module is available.

1.6.2. <u>Post-Conditions</u>

Background noise is minimized from the audio file. The cleaned audio is stored temporarily. The audio is ready for the next processing step.

1.7. ***Converting Audio Format (Medium Priority)*** | 3-point story

From Process Audio Epic. As a musician, I want my recorded or uploaded audio to be converted into a compatible format so that it can be analyzed correctly for pitch detection.

1.7.1. Pre-Conditions

A valid audio file exists (recorded or uploaded). The audio processing module is active.

1.7.2. Post-Conditions

The audio file is converted into a standardized format (e.g., WAV, mono, fixed sample rate). The converted file replaces or updates the previous version in session storage. The file is ready for normalization.

1.8. Normalizing Sound Levels (Medium Priority) | 5-point story

From Process Audio Epic. As a musician, I want the sound levels of my recording to be normalized so that pitch detection is not affected by uneven volume levels.

1.8.1. Pre-Conditions

A valid MP3 audio recording file exists. The selected instrument settings are loaded. The audio processing module is active.

1.8.2. Post-Conditions

The audio is cleaned and normalized. The audio is converted into a format ready for pitch detection. The processed audio is stored in the system.

1.9. Detect Pitch (High Priority) | 3-point story

As the system, I will ensure pitch is accurate so that the exact frequency and letter notation can be determined.

1.9.1. Pre-Conditions

Processed audio is available and the correct instrument settings are loaded.

1.9.2. Post-Conditions

The fundamental frequency in hertz (Hz) is found. Detected pitch values are stored.

1.10. Identify Letter Notation (Medium Priority) | 2-point story

As the system, I will identify the letter notation so that the musician can confirm the notes they are playing are the correct ones.

1.10.1. Pre-Conditions

Pitch frequency was successfully detected. The pitch-to-note mapping is available, with a reference table available as well.

1.10.2. *Post-Conditions*

The detected frequency is mapped to the closest note/letter and stored. The system saves the note information for evaluation.

1.11. ***Evaluate Pitch Accuracy (High Priority)*** | 3-point story

As the system, I will evaluate pitch accuracy/intonation and give feedback on whether the note was played in tune, sharp, or flat.

1.11.1. *Pre-Conditions*

The note has been identified. The detected and expected pitch frequencies are available. Related information and rules regarding the selected instrument are also loaded.

1.11.2. *Post-Conditions*

The pitch is evaluated as in tune, sharp, or flat. The deviation from the proper pitch is calculated. The result is stored and out-of-tune pitches are flagged.

1.12. ***Validate Performance (High Priority)*** | 3-point story

As the system, I want to validate the audio recording so the user can confirm they are following the on-screen instructions/guidance/exercise.

1.12.1. *Pre-Conditions*

Pitch accuracy has been evaluated and the selected instrument is set.

1.12.2. *Post-Conditions*

The detected pitch is validated and the system determines if the pitch is within an acceptable deviation. Wrong notes are flagged, and the results are stored.

1.13. ***Send data to AI (High Priority)*** | 3-point story

From Generate Feedback Epic. As the system, I want to send all audio data from the file to the Groq API to analyze the letter notation depending on the exercise.

1.13.1. Pre-Conditions

The audio file has been uploaded and processed by the web application.

1.13.2. Post-Conditions

Training and data analysis has been completed by the Groq API and is ready to send back to the web application. Data is ready to use for a feedback message.

1.14. Generate feedback message (High Priority) | 5-point story

From Generate Feedback Epic. As the system, I want to generate a personalized feedback message based on the Groq API to display to the musician using the service.

1.14.1. Pre-Conditions

The audio file is successfully processed by Groq API.

1.14.2. Post-Conditions

Feedback is successfully generated and is ready to be sent to the web application to display to the musician.

1.15. Display generated feedback (Medium Priority) | 2-point story

From Generate Feedback Epic. As the system, I want to display the generated feedback to the musician to help them improve their performance and future recordings.

1.15.1. Pre-Conditions

Performance validation and pitch evaluation results are available in the system. The Groq API connection and AI feedback modules are active.

1.15.2. Post-Conditions

Feedback text is generated, stored, and sent to the Audiophile web application. Any errors from the API are logged and handled.

1.16. View Feedback Results (Medium Priority) | 2-point story

As a musician, I want clear access to personalized feedback so that I can effectively improve my skillset, knowledge, and performance.

1.16.1. Pre-Conditions

Feedback has been generated successfully. Evaluation results are available as well.

1.16.2. Post-Conditions

Feedback is displayed on the web application with results for pitch and note accuracy visible. The user can review the results and the session remains active to be reused if needed.

2. User Story Size Summary

Story ID	User Story	Size	Priority
1	Select Instrument	1	Low
2	Start Recording Audio	2	High
3	Stop Recording Audio	2	High
4	Save Recorded Audio	2	Medium
5	Upload Audio File	1	Medium
6	Reduce Background Noise	5	High
7	Convert Audio Format	3	Medium
8	Normalize Sound Levels	5	Medium
9	Detect Pitch Frequency	3	High
10	Identify Letter Notation	2	Medium
11	Evaluate Pitch Accuracy	3	High
12	Validate Performance	3	High
13	Send Data to Groq API	3	High
14	Generate Feedback Message	5	High
15	Display Generated Feedback	2	Medium
16	View Feedback Results Page	2	Medium
Totals	**-**	**44**	**-**

3. Sprint 1 Subset, Audio Input and Preprocessing

Story ID	User Story	Size	Priority
1	Select Instrument	1	Low
2	Start Recording Audio	2	High
3	Stop Recording Audio	2	High
4	Save Recorded Audio	2	Medium
5	Upload Audio File	1	Medium
6	Reduce Background Noise	5	High
7	Convert Audio Format	3	Medium
Totals	**–**	**18**	**–**

4. Functionality at End Of Sprint 1

4.1. Sprint 1 Functionality Description

At the end of Sprint 1, the system will support the core audio input and initial processing pipeline. The user will be able to open the web application and select their instrument from a list of supported options, which configures the session for instrument-specific analysis later in the program.

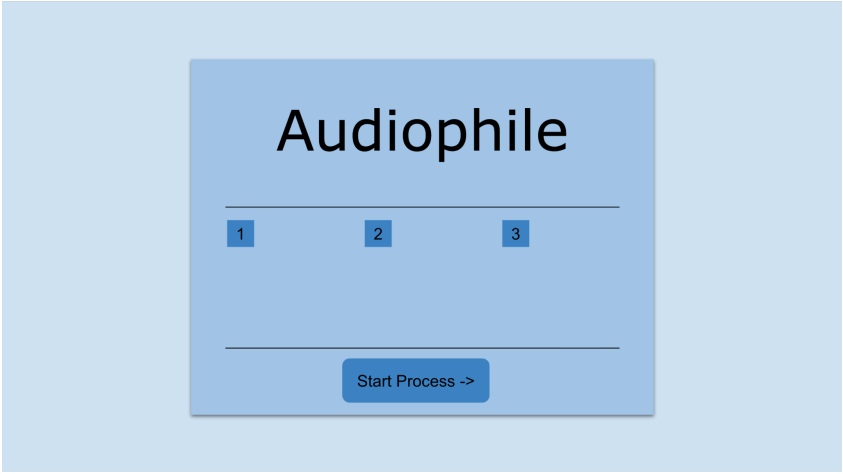
Once an instrument is selected, the user can navigate to the recording interface and start capturing live audio through their microphone. The interface will display a visual indicator confirming that recording is active. The user can then stop the recording at any time, at which point the audio is finalized and saved temporarily within the system session. The user will receive a confirmation that their recording has been saved and is ready for processing.

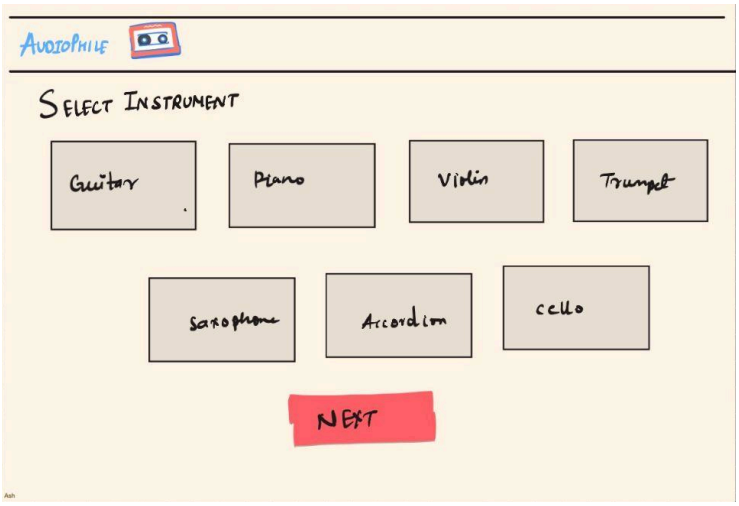
Following the save, the system will automatically begin the first stages of audio processing. Background noise will be reduced to minimize the impact of external sounds on later pitch detection. The audio will then be converted into a standardized format compatible with the pitch detection module.

By the end of Sprint 1, the system will not yet detect pitch, evaluate accuracy, or generate feedback. However, it will have established a complete and functional audio input pipeline, from instrument selection through to a cleaned and formatted audio file, providing a stable foundation for the rest of the program.

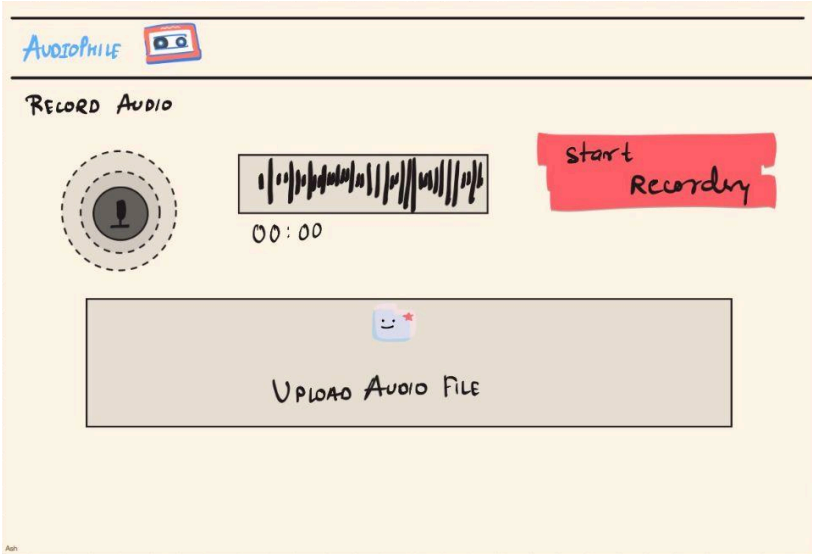
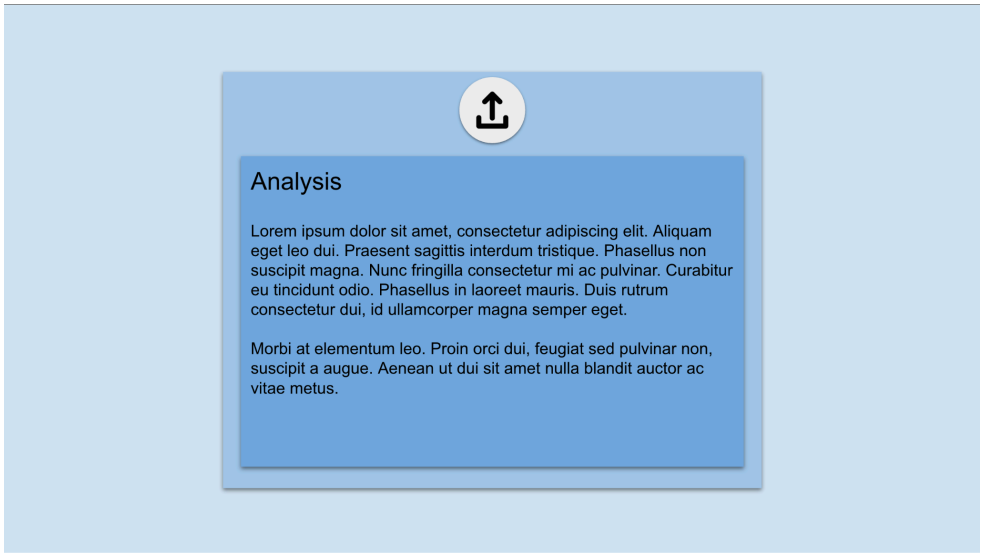
5. Key Features of the UI

5.1. Landing Page





5.2. Record / Upload Selection Page



5.3. Analysis Page